Tandle Vardhan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.24
Sri Chaitanya Junior College	Class XII (Telangana State Board of Intermediate Education)	2020	8.64%
English Union High School	Class X (Telangana State Board of Secondary Education)	2018	9.50%

PROJECTS

CNN Model Compression & Edge Deployment | *PyTorch, ONNX Runtime, FastAPI*

- Built a compression pipeline for ResNet18 on CIFAR-100: INT8 post-training quantization cut model size 4× and CPU inference latency ~1.8× with negligible accuracy loss (83.6% 83.5%).
- Applied 50% structured filter pruning with fine-tuning; combined with INT8 for a 7.7× smaller, ~3× faster model at 80.1% accuracy.
- Exported all variants to ONNX and served the compressed model via a FastAPI inference endpoint.

Bill Tracker Agent | *Python, Gemini API, Gmail API, Google Sheets API, FastAPI, Docker, Google Cloud Run*

- Built an LLM agent that uses ReAct: a reason-act-observe loop where the model uses tools to read a Gmail inbox (read-only), check each extracted bill, log it to Google Sheets, and flag anything overdue or more than 20% above a vendor's usual charge.
- Enforced guardrails in code: a grounding check rejects any amount or vendor not found in the source email, and a filter blocks prompt-injection emails; handled 7 of 7 injection test cases.
- Evaluated on 66 labelled emails including adversarial cases: 98% fully correct (65/66), overdue-flag precision 1.00 and recall 0.93, at 5 LLM calls and about 39k tokens per 22-email run; documented the failure and a guardrail bug the eval exposed. Deployed and tested as a Dockerised FastAPI service on Google Cloud Run

AskPage - Browser Q&A Assistant with Citations | *Sentence-Transformers, Ollama, FAISS, BM25, FastAPI, Docker*

- Built a Chrome extension side panel for on-page Q&A, backed by a Dockerized FastAPI RAG server streaming answers from a local LLaMA 3.2 model.
- Designed a hybrid retrieval pipeline combining FAISS, BM25, Reciprocal Rank Fusion, and MiniLM cross-encoder re-ranking, raising hit@1 from 0.63 to 0.84 and MRR from 0.72 to 0.92 at 143 ms median CPU latency.
- Implemented section-level chunking so every claim carries a clickable citation, with 97% of answers cited; tested on 40 questions across 4 live websites, including 8 the pages could not answer, to check the model declines instead of guessing.

Mini-GPT -- Decoder-Only Transformer from Scratch | *PyTorch, tiktoken, Gradio*

- Built a GPT-style decoder-only Transformer from scratch in PyTorch -- multi-head causal self-attention, pre-LayerNorm residual blocks, GELU feed-forward layers, GPT-2 BPE tokenization (tiktoken), and learned positional embeddings -- trained on next-token prediction over Tiny Shakespeare.
- Implemented **KV-cache** inference with **temperature, top-k, and top-p sampling**, and served the model via a Gradio web app.

SKILLS

Programming Languages: Python, C++, SQL

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn

Libraries & Data Processing: NumPy, Pandas, SciPy, OpenCV, Matplotlib, Seaborn

GenAI & LLMs: Transformers, RAG, LangChain, LangGraph, Hugging Face, FAISS, Ollama, ONNX Runtime

Tools: Git, Docker, FastAPI, Google Cloud Run, Jupyter

RELEVANT COURSES

Core AI & Data Science: Foundations of Machine Learning, Deep Learning, Visual Computing, Fraud Analytics, Generative AI.

CS Fundamentals: Introduction to Programming, Data Structures & Algorithms.

Mathematics & Theory: Linear Algebra, Probability, Statistics, Calculus, Convex Optimization, Matrix Theory,

POSITIONS OF RESPONSIBILITY

Internship Coordinator: Acted as the primary single point of contact between students and hiring managers to facilitate seamless internship hiring processes.

Teaching Assistant (Multi-Agent Systems): Streamlined grading and feedback for assignments submitted by 133 students.

Design Coordinator (Milan Inter-Hostel Championship): Delivered visual assets, including posters, jerseys, and props, using Adobe Photoshop to elevate event branding.

EXTRACURRICULAR

Fine arts -- sketching, painting (A+ in Madhubani painting course).

Image Editing -- 2nd prize, image editing competition, Elan & nVision (IIT Hyderabad's annual techno-cultural fest)